

Moosa Abbasi

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Education

University of South Florida, Honors College

Bachelor of Science in Computer Science

GPA: 3.90/4.00 | Green and Gold Presidential Scholar, Dean's List

Tampa, FL

Aug. 2023 – Aug. 2027

Relevant Coursework: Data Structures, Analysis of Algorithms, Computer Architecture, Software Engineering, Computer Logic & Design, Automata Theory, Introduction to Artificial Intelligence, Discrete Structures, Honors Capstone

Technical Skills

Languages & Databases: Python, C, C++, JavaScript, SQL, DynamoDB, MySQL

Frameworks & Libraries: Streamlit, React.js, Flask, OpenCV, Pandas, NumPy, Plotly, scikit-learn, XGBoost, Prophet

Cloud & DevOps: AWS (Lambda, API Gateway, S3, IAM), Google Cloud (Cloud Run, Secret Manager, GCS), Docker, Git, GitHub Actions

Security & Testing: Burp Suite, OWASP ZAP, OWASP WSTG v4.2, Playwright, HTTP Traffic Analysis, XSS/SQLi, Prompt Injection

Certifications: Machine Learning Specialization (Supervised ML, Advanced Learning Algorithms, Unsupervised Learning): Andrew Ng, DeepLearning.AI & Stanford University, Coursera · 2026

Experience

Undergraduate Research Assistant: AI Course Companion (AICC)

May 2026 – Present

University of South Florida | Dr. Oguzhan Topsakal

- Built an automated security-testing suite (**Python + Playwright**) assessing AICC across all **OWASP WSTG v4.2** categories plus **LLM prompt-injection testing**, with a **Gemini-based evaluator** to cut false positives against the LLM backend.
- Ran **black-box then gray-box source review** of the **OAuth flow** and backend **API**, tracing authentication and data-serving logic across the codebase, and compiled an **OWASP-mapped, severity-rated** report for the faculty supervisor.
- Containerized the suite with **Docker** and deployed it to **Google Cloud Run** (Job + private **Cloud Run Service UI**) using **Secret Manager, Cloud Storage, and IAM**, a fully cloud-hosted platform now used for ongoing AICC security testing.

Project Management Intern

May 2024 – Aug 2024

Huawei Technologies

- Resolved visibility gaps across **5+ concurrent DMD projects** by building structured **tracking dashboards** that standardized reporting cycles and cross-team workflows, improving on-time delivery by **20%**.
- Eliminated siloed communication by introducing a **data-driven reporting** system and standardized **project documentation** workflow, reducing delays by **25%** and boosting team efficiency by **15%**.

Projects

Smart Job Tracker

May 2025 – Present

AWS Lambda, DynamoDB, API Gateway, Cognito, SES, Chrome Extension (MV3), Streamlit, Python

Tampa, FL

- Eliminated manual logging via a **Chrome Extension (MV3)** with a **3-layer detection system** using **MAIN world XHR monkey-patching** and **MutationObserver**, plus a **chrome.alarms** queue guaranteeing **zero application loss** on service worker suspension.
- Engineered a **5-Lambda serverless backend** with **DynamoDB user-scoped isolation**, **API Gateway REST API** protected by a **Cognito JWT** authorizer, **preemptive token refresh** with race-condition locking, and automatic **401 retry** across all 3 clients.
- Deployed a live **Streamlit** dashboard with **EventBridge + SES** daily email reminders for application tracking; reduced total operating cost to **\$0/month** by architecting entirely within the **AWS Free Tier**.

USF Emissions Analysis Dashboard

Jan 2026 – Present

Python, Streamlit, Plotly, Pandas, scikit-learn, XGBoost, Prophet, GHG Protocol

Tampa, FL

- Quantified **~83.8M kg CO₂** across **5 categories** (Scope 1 + Scope 3 Cat 4/6/7/9) by building a full-stack **GHG emissions dashboard** in **Python/Streamlit**, applying **GHG Protocol** and **EPA SmartWay** emission factors to **~50,000 synthetic records**.
- Built **3 ML models: IsolationForest** anomaly detection flagging **500 shipments** with **6.3M kg CO₂** savings potential, **Prophet** 24-month emissions forecasting, and **XGBoost** regression at **R²=0.9965** with invoice amount as the top emission predictor.
- Deployed on **Streamlit Cloud** via **GitHub** with automated redeployment; presented findings to an airport industry client in collaboration with **Patel's College of Global Sustainability** for Scope 1 and Scope 3 strategy recommendations.

MOPS — Class Scheduling System

Jan 2026 – Apr 2026

Python, FastAPI, SQLAlchemy, React, TypeScript, SQLite, Tailwind

Tampa, FL

- Built the **full-stack** MOPS scheduling system end-to-end in a **4-person Agile team**: a **FastAPI + SQLAlchemy + Pydantic** backend over **SQLite** with a **7-layer** service/repository architecture and a versioned **REST API** of **10+** endpoints documented via **Swagger**.
- Developed the **React + TypeScript** frontend (**Vite, Tailwind**) delivering **section management**, a **room-utilization heat map**, and cross-semester **enrollment analytics** with near-capacity and **growth indicators** across **3 semesters** of college data.
- Engineered a **3-check audit engine** (duplicate CRNs, time conflicts, missing fields), **2-role access control** (Member/Chair), **schedule lock/unlock** with modification history, and **PDF export** for **schedules, audit reports, and enrollment comparisons**.